

Lecture 4

average instantaneous Power

$$S_{av} = P = \frac{(E_0)^2}{2Z_0} \cos \theta$$

$\leftarrow$ Watts
 $\leftarrow$ Peak amplitude
 $\theta \rightarrow$ temporal phase shift between $\vec{E}$ and $\vec{H}$
 Power over space ($\frac{W}{m^2}$)

Similarities * P looks like $\frac{(V_{peak})^2}{2Z}$ from Power

$$\tilde{I}(\text{irradiance}) = P = \frac{E_0^2}{2Z_0}$$

if not in free space $\downarrow$ free space permeability

$$Z = \sqrt{\frac{\mu}{\epsilon}} \quad ; \quad \mu = \mu_0 \mu_r \leftarrow \text{relative permeability}$$

(ratio, dimensionless) $\mu_r = 1$ for free space

$$\epsilon = \epsilon_0 \epsilon_r \leftarrow \text{relative permittivity}$$

$\uparrow$
Permittivity

limit ourselves to dielectric materials $\rightarrow (\sigma = 0; \mu_r = 1)$ assumptions

$$Z = \sqrt{\frac{\mu_0}{\epsilon_0 \epsilon_r}} \rightarrow \text{using our assumptions}$$

$$\tilde{I} = \frac{E_0^2}{2 \sqrt{\frac{\mu_0}{\epsilon_0 \epsilon_r}}} \rightarrow \frac{E_0^2 \sqrt{\epsilon_0 \epsilon_r}}{2 \sqrt{\mu_0}} \left(\frac{\sqrt{\epsilon_0}}{\sqrt{\epsilon_0}} \right) \rightarrow \frac{E_0^2 \epsilon_0 \sqrt{\epsilon_r}}{2 \sqrt{\mu_0 \epsilon_0}} = \boxed{\frac{(E_0)^2 c \epsilon_0 \sqrt{\epsilon_r}}{2} \left[\frac{W}{m^2} \right]}$$

$$v = \frac{1}{\sqrt{\epsilon \mu}} \rightarrow \frac{1}{\sqrt{\epsilon_0 \epsilon_r \mu_0}} = \frac{c}{\sqrt{\epsilon_r}}$$

* $\epsilon_r \geq 1$ in any material

$c = f \lambda ; \boxed{\vec{v} = \frac{f \lambda}{\sqrt{\epsilon_r}}}$

$\downarrow \frac{1}{\sqrt{\mu_0 \epsilon_0}} = c$

$\sqrt{\epsilon_r} = n \rightarrow$ Refractive index or Index of Refraction (n)

$$n = \frac{c}{v} \quad \text{velocity}$$

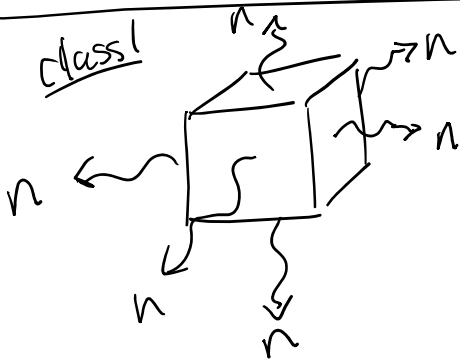
$$\mathcal{I} = \frac{\epsilon_0 n c E_0^2}{2}$$

$k_0 \rightarrow$ Propagation constant in freespace

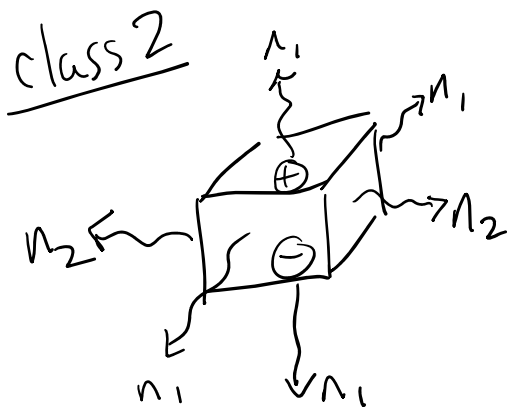
$$k_0 = \frac{2\pi}{\lambda_0} \rightarrow k(\text{medium}) = n k_0$$

$$\lambda(\text{medium}) = \frac{\lambda_0}{n}$$

Define 2- classes of materials



* n is the same in all directions
a wave would propagate
(isotropic)
• most things are like this



* n is not the same in all
directions
(anisotropic)

• quartz (piezo-electric)

• velocity is different
in different direction

if n varies then the frequency (aka the speed) will vary

$$f = \frac{c}{n\lambda} \rightarrow \boxed{v = f n \lambda}$$

* and the variation will be based on freq.

* Prisms: things that separate light.

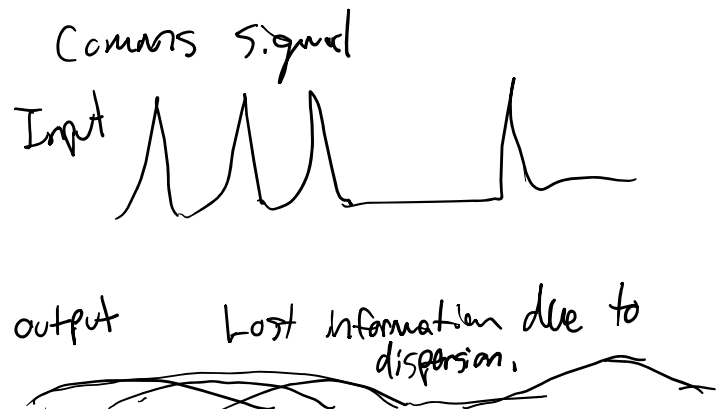
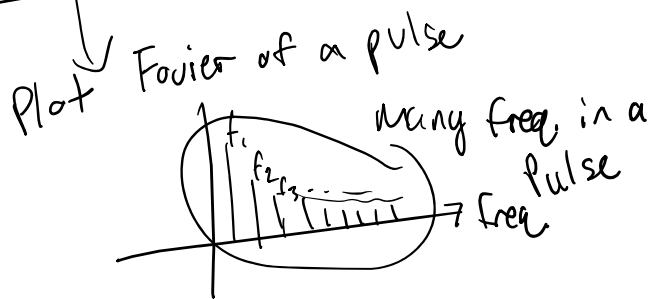
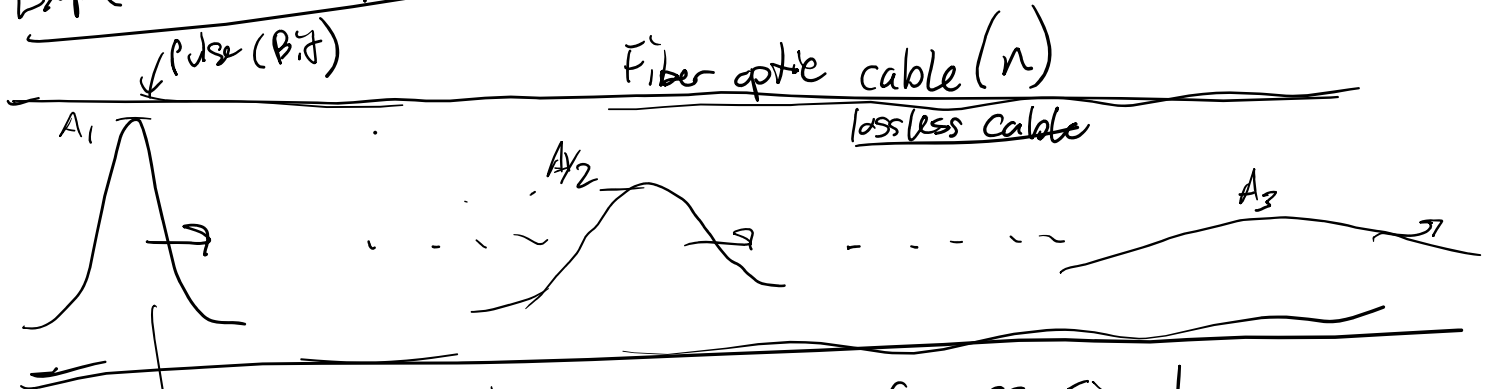
* can be bad in fiber optics

$$\lambda_{\text{medium}} = \frac{\lambda_0}{n}; n \rightarrow \text{Refractive index}$$

Remember

* when the speed of a wave thru a material depends on its freq. $\rightarrow$ called Dispersion.

Exp (bad example)



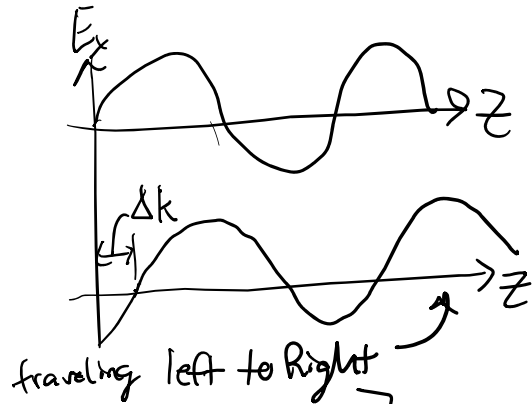
*lets Look at math

Assume we have two waves of slightly different freq. ($\Delta\omega$) and the propagate in a medium with dispersion, resulting in a Δk .

Wave 1 $\rightarrow \omega - \Delta\omega ; k - \Delta k$

Wave 2 $\rightarrow \omega + \Delta\omega ; k + \Delta k$

* Assume $E_{o1} = E_{o2} = E_o$



Wave 1

$$E_{x1}(z,t) = E_o \cos \left[(\omega - \Delta\omega)t - (k - \Delta k)z \right]$$

Wave 2

$$E_{x2}(z,t) = E_o \cos \left[(\omega + \Delta\omega)t - (k + \Delta k)z \right]$$

Waves are traveling in the same direction

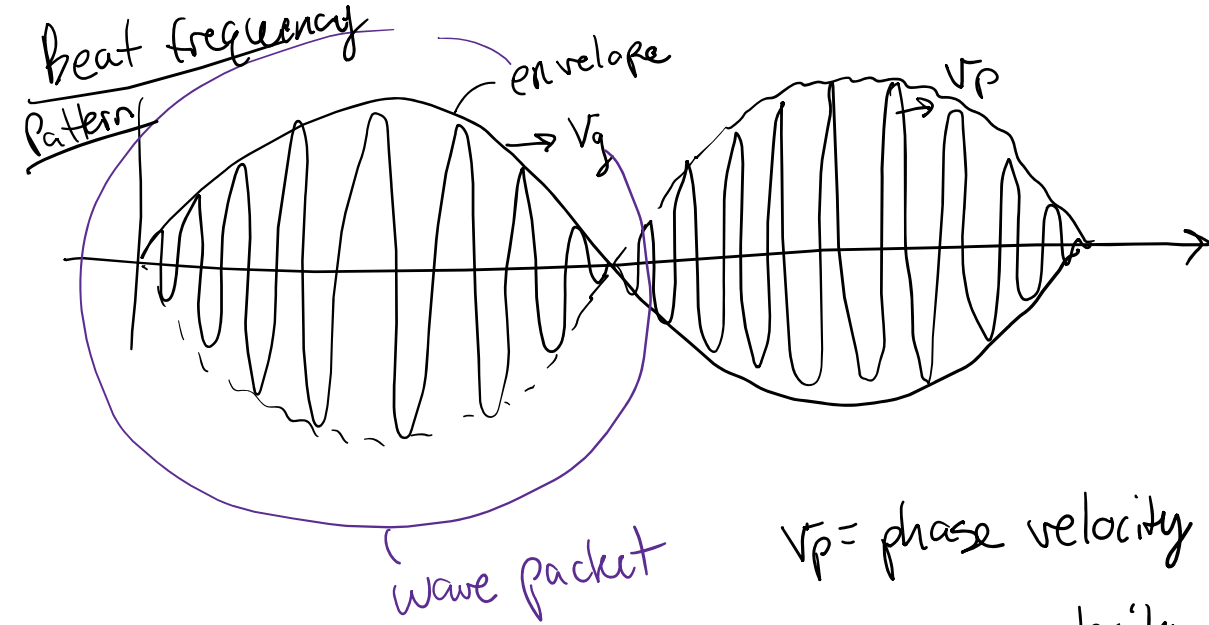
The total field in the medium

$$E_{xT}(z,t) = E_o \cos \left[\underbrace{(\omega - \Delta\omega)t - (k - \Delta k)z}_A \right] + E_o \cos \left[\underbrace{(\omega + \Delta\omega)t - (k + \Delta k)z}_B \right]$$

$$= E_o \left(\cos A + \cos B \right) \Rightarrow E_o \left(2 \cos \left(\frac{1}{2}(A+B) \right) \cos \left(\frac{1}{2}(A-B) \right) \right)$$

working

$$E_{xT}(z,t) = 2E_o \cos \left[(\Delta\omega)t - (\Delta k)z \right] \cos(\omega t - kz)$$



v_p = phase velocity

v_g = group velocity